

Day 27: Enterprise Architecture & Semantic Models

[DEF] Definition to Remember

Enterprise Power BI Architecture decouples data modeling from report visualization through Golden Semantic Models (Hub-and-Spoke), User Aggregations, and Deployment Pipelines.

1. Hub-and-Spoke Architecture (Thin Reports)

- * Anti-Pattern (Siloed): 20 separate .pbix files, each connecting to SQL and duplicating ETL and DAX logic.

[Golden Semantic Model] (Single Source of Truth)

[Sales Report] [Finance Report] [Operations Report]

2. User-Defined Aggregations

- * Pair a massive DirectQuery fact table (billions of rows) with an in-memory Import aggregation table (summarized by month/region).

3. Deployment Pipelines (Dev -> Test -> Prod)

- * Automated lifecycle management promoting reports and datasets across Development, Staging/Test, and Production workspaces.

[MUST-WRITE] Must-Master Skills & Interview Points

1. Benefits of the Hub-and-Spoke architecture for enterprise governance and metric consistency.
2. How User-Defined Aggregations deliver in-memory speed over big data lakes.
3. Implementing Dev/Test/Prod Deployment Pipelines in Power BI Service.

[RECALL] Quick Recall

Golden Semantic Model (Single Source of Truth) + Thin Reports + Deployment Pipelines (Dev -> Test -> Prod)